

## SISKIYOU COUNTY AP, CA, USA

WMO: 725955

Lat: 41.781N

Lon: 122.468W

Elev: 808

StdP: 91.99

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24259

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 12            | -8.6       | -7.0 | -13.3                         | 1.3 | -3.8 | -10.9 | 1.6 | -2.2 | 15.5                  | 7.6  | 14.1 | 7.5  | 0.8                   | 30   | 0.561 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 19.6                   | 36.2            | 17.6 | 34.4 | 17.0 | 32.7 | 16.4 | 18.9                | 32.9 | 18.0 | 31.9 | 17.2 | 30.8 | 3.4                  | 0    |

| Dehumidification DP/MCDB and HR |      |      |      |     |      |      |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|-----|------|------|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |     |      | 2%   |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR  | MCDB | DP   | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e) | (f)  | (g)  | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 13.7                            | 10.8 | 22.3 | 12.4 | 9.9 | 22.3 | 11.4 | 9.2 | 21.9 | 56.5          | 33.1 | 53.6 | 31.9 | 51.0 | 30.9 | 23.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
| 11.9              | 9.7  | 8.2 | -12.9                      | 39.7 | 4.2                | 1.7 | -16.0                                              | 40.9 | -18.4      | 41.8 | -20.8      | 42.8 | -23.8      | 44.0 |  |  |
|                   |      |     | -13.4                      | 20.5 | 4.1                | 1.1 | -16.3                                              | 21.3 | -18.7      | 22.0 | -21.0      | 22.6 | -24.0      | 23.4 |  |  |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 11.3   | 2.5   | 4.2   | 6.5   | 9.2   | 13.5  | 17.9  | 23.1  | 22.1  | 17.9  | 11.2  | 5.4   | 1.8  |
|                                                                                  | DBStd       | 8.06   | 3.37  | 3.17  | 3.20  | 3.71  | 3.95  | 4.06  | 2.99  | 2.67  | 3.60  | 3.59  | 3.68  | 3.94 |
|                                                                                  | HDD10.0     | 1009   | 234   | 163   | 113   | 59    | 12    | 1     | 0     | 0     | 1     | 27    | 146   | 253  |
|                                                                                  | HDD18.3     | 2914   | 492   | 396   | 366   | 273   | 155   | 55    | 3     | 3     | 51    | 221   | 389   | 511  |
|                                                                                  | CDD10.0     | 1496   | 1     | 1     | 7     | 36    | 122   | 239   | 407   | 374   | 238   | 66    | 6     | 1    |
|                                                                                  | CDD18.3     | 359    | 0     | 0     | 0     | 0     | 7     | 43    | 152   | 120   | 37    | 1     | 0     | 0    |
|                                                                                  | CDH23.3     | 6217   | 0     | 0     | 1     | 37    | 253   | 856   | 2257  | 1877  | 862   | 74    | 0     | 0    |
|                                                                                  | CDH26.7     | 3046   | 0     | 0     | 0     | 4     | 78    | 371   | 1231  | 969   | 377   | 15    | 0     | 0    |
| Wind                                                                             | WSAvg       | 2.7    | 2.3   | 2.8   | 3.1   | 3.0   | 3.0   | 3.3   | 3.0   | 2.7   | 2.4   | 2.3   | 2.2   | 2.4  |
| Precipitation                                                                    | PrecAvg     | 486    | 78    | 52    | 51    | 38    | 36    | 21    | 11    | 8     | 9     | 34    | 62    | 85   |
|                                                                                  | PrecMax     | 782    | 158   | 128   | 206   | 89    | 95    | 61    | 43    | 36    | 29    | 122   | 197   | 225  |
|                                                                                  | PrecMin     | 276    | 18    | 24    | 13    | 16    | 9     | 0     | 0     | 0     | 0     | 4     | 10    | 16   |
|                                                                                  | PrecStd     | 131    | 45    | 29    | 40    | 19    | 23    | 17    | 11    | 8     | 7     | 29    | 41    | 60   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 13.9  | 17.5  | 22.0  | 27.1  | 32.1  | 36.1  | 38.9  | 37.9  | 34.7  | 28.7  | 19.9  | 13.2 |
|                                                                                  |             | MCWB   | 8.1   | 9.8   | 11.1  | 13.8  | 15.8  | 17.6  | 18.9  | 17.9  | 16.0  | 14.4  | 11.2  | 9.0  |
|                                                                                  | 2%          | DB     | 11.7  | 14.7  | 19.0  | 23.9  | 28.8  | 33.0  | 36.9  | 35.7  | 32.7  | 25.5  | 16.7  | 11.2 |
|                                                                                  |             | MCWB   | 7.3   | 8.0   | 10.0  | 12.7  | 15.0  | 16.5  | 18.0  | 17.2  | 15.4  | 12.8  | 9.9   | 7.6  |
|                                                                                  | 5%          | DB     | 9.8   | 12.4  | 16.7  | 21.1  | 26.2  | 31.0  | 35.0  | 33.8  | 31.1  | 22.7  | 14.1  | 9.7  |
|                                                                                  |             | MCWB   | 6.3   | 7.1   | 9.0   | 11.3  | 13.9  | 15.7  | 17.5  | 16.5  | 15.1  | 11.8  | 8.6   | 6.5  |
|                                                                                  | 10%         | DB     | 8.1   | 10.5  | 13.9  | 17.9  | 23.4  | 28.6  | 33.0  | 32.2  | 28.8  | 20.2  | 12.1  | 8.1  |
|                                                                                  |             | MCWB   | 5.3   | 6.0   | 7.7   | 9.9   | 12.9  | 14.9  | 17.0  | 16.1  | 14.4  | 10.7  | 7.7   | 5.4  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 9.4   | 10.5  | 12.4  | 14.9  | 17.3  | 19.0  | 20.5  | 19.4  | 17.8  | 15.1  | 12.5  | 9.7  |
|                                                                                  |             | MCDB   | 12.1  | 15.9  | 19.7  | 24.8  | 29.2  | 32.8  | 35.0  | 34.2  | 32.2  | 25.7  | 18.1  | 12.4 |
|                                                                                  | 2%          | WB     | 7.8   | 9.0   | 10.8  | 13.1  | 15.7  | 17.5  | 19.4  | 18.3  | 16.4  | 14.0  | 10.7  | 8.0  |
|                                                                                  |             | MCDB   | 10.6  | 13.6  | 17.5  | 22.5  | 26.8  | 30.9  | 33.3  | 32.6  | 30.0  | 24.0  | 15.5  | 10.4 |
|                                                                                  | 5%          | WB     | 6.7   | 7.6   | 9.6   | 11.7  | 14.7  | 16.3  | 18.5  | 17.5  | 15.5  | 12.6  | 9.4   | 6.9  |
|                                                                                  |             | MCDB   | 9.3   | 11.6  | 15.3  | 20.0  | 24.7  | 29.0  | 32.3  | 31.8  | 29.1  | 21.1  | 13.4  | 9.2  |
|                                                                                  | 10%         | WB     | 5.6   | 6.4   | 8.3   | 10.3  | 13.4  | 15.3  | 17.7  | 16.7  | 14.7  | 11.3  | 8.0   | 5.7  |
|                                                                                  |             | MCDB   | 8.0   | 10.1  | 13.5  | 17.2  | 22.4  | 27.1  | 31.3  | 30.4  | 27.7  | 18.7  | 11.6  | 7.8  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 9.7   | 11.9  | 13.4  | 14.7  | 16.0  | 17.9  | 19.6  | 19.9  | 19.8  | 16.7  | 11.8  | 9.4  |
|                                                                                  |             | MCDBR  | 12.3  | 15.5  | 19.0  | 20.9  | 21.3  | 21.7  | 21.6  | 22.2  | 23.4  | 21.4  | 15.4  | 10.0 |
|                                                                                  |             | MCWBR  | 8.4   | 9.7   | 10.9  | 11.0  | 9.6   | 8.4   | 8.0   | 8.6   | 10.5  | 11.3  | 9.8   | 6.9  |
|                                                                                  | 5% WB       | MCDBR  | 10.5  | 14.1  | 16.7  | 19.6  | 19.6  | 20.4  | 20.2  | 20.6  | 21.9  | 19.0  | 13.6  | 8.7  |
|                                                                                  |             | MCWBR  | 7.4   | 9.1   | 9.8   | 10.7  | 9.1   | 8.2   | 7.6   | 8.1   | 10.2  | 10.3  | 8.8   | 6.2  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.291  | 0.295 | 0.301 | 0.319 | 0.326 | 0.319 | 0.324 | 0.320 | 0.302 | 0.301 | 0.286 | 0.295 |      |
|                                                                                  | taud        | 2.531  | 2.545 | 2.554 | 2.494 | 2.513 | 2.566 | 2.563 | 2.555 | 2.592 | 2.592 | 2.598 | 2.516 |      |
|                                                                                  | Ebn at Noon | 870    | 924   | 955   | 953   | 948   | 951   | 943   | 940   | 941   | 904   | 874   | 831   |      |
|                                                                                  | Edh at Noon | 71     | 81    | 90    | 103   | 104   | 99    | 98    | 96    | 86    | 76    | 65    | 66    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.87   | 2.77  | 3.92  | 5.38  | 6.74  | 7.83  | 8.04  | 7.11  | 5.60  | 3.68  | 2.14  | 1.53  |      |
|                                                                                  | RadStd      | 0.29   | 0.35  | 0.41  | 0.53  | 0.53  | 0.44  | 0.29  | 0.30  | 0.27  | 0.35  | 0.25  | 0.30  |      |

## Historical Trends

|                         |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                         |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| Station Only            |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
|                         |  | +0.48 | N/A     | N/A    | +0.63   | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| Regional (16 neighbors) |  | N/A   | N/A     | N/A    | +0.52   | N/A   | N/A   | N/A         | N/A     | +98     | +18     |

Nomenclature: See separate page